

Part I

In this part we perform Gaussian elimination without pivoting to obtain the **L-U** factorization of a 4×4 matrix.

Create the matrix

$$\mathbf{A} = \begin{bmatrix} 3 & 2 & -1 & 2 \\ -3 & -4 & 2 & 1 \\ 6 & 2 & -2 & 11 \\ -6 & -10 & 6 & 2 \end{bmatrix}$$

1. Initialize the matrix **L**.

```
>>L = eye(4)
```

2. Compute the multipliers and eliminate the entries of **A** in the first column. Observe that the matrix **L** is being created and the matrix **A** is being over-written in the process. Make sure you understand the commands.

```
>>L(2,1) = A(2,1)/A(1,1)
>>A(2,1:4) = A(2,1:4)-L(2,1)*A(1,1:4)
>>L(3,1)=A(3,1)/A(1,1)
>>A(3,1:4)=A(3,1:4)-L(3,1)*A(1,1:4)
>>L(4,1)=A(4,1)/A(1,1)
>>A(4,1:4)=A(4,1:4)-L(4,1)*A(1,1:4)
```

3. Compute the multipliers and eliminate the entries of **A** in the second column.

```
>>L(3,2)=A(3,2)/A(2,2)
>>A(3,2:4)=A(3,2:4)-L(3,2)*A(2,2:4)
>>L(4,2)=A(4,2)/A(2,2)
>>A(4,2:4)=A(4,2:4)-L(4,2)*A(2,2:4)
```

4. Compute the multipliers and eliminate the entries of **A** in the third column.

```
>>L(4,3)=A(4,3)/A(3,3)
>>A(4,3:4)=A(4,3:4)-L(4,3)*A(3,3:4)
```

5. The matrix **A** has been transformed to an upper triangular matrix. So we rename it **U**.

```
>>U=A
>>L
>>A=L*U
```

You should recover the original matrix **A**.

The commands above can be simplified by using **for** loops. Moreover, we can store the commands in an M-file.

Use an M-file window to create the following file, which starts with three comment lines:

```
% The function in this M-file computes the L-U factorization of a  $4 \times 4$  matrix  
% without pivoting. To use this function, create a  $4 \times 4$  matrix A and type  
% [L,U]=LU4(A).
```

```
function [L,U]=LU4(A)  
L=eye(4);  
for j=1:3  
    for i=j+1:4  
        L(i,j)=A(i,j)/A(j,j);  
        A(i,j:4)=A(i,j:4)-L(i,j)*A(j,j:4);  
    end  
end  
U=A;
```

Save this file as LU4.m

Now go back to your MATLAB window and type the following commands.

```
>>clear L U
```

```
>>help LU4
```

What appeared on the screen?

```
>>[L,U]=LU4(A)
```

The previously obtained **L** and **U** should reappear.

Clear all the variables before you work on Part II.